SSIS to Azure Migration Assessment Engine

System Architecture & Automated Evaluation Framework

This technical specification outlines the design of a rule-based and LLM-augmented parser designed to scan SSIS `.dtsx` packages. The tool inventories control flows, data flows, connections, and script complexities to route each asset to its optimal Azure target destination.

1. Target Architecture Routing Matrix

The evaluation framework checks packages against specific architectural markers to map them to three primary cloud destinations:

SSIS Component Profile	Target Recommendation	Migration Complexity	Justification
Standard sources/ sinks, basic OLE DB destinations, Execute SQL Tasks, basic Derived Columns.	**Native ADF Pipeline**	**Low**	Direct mapping available using ADF Copy Activities, Lookups, and native Mapping Data Flows. High optimization, low cost.
Extensive custom C# or VB.NET Script Tasks, third-party proprietary components (Attunity, CozyRoc), or reliance on local Excel/ Access drivers.	**Azure-SSIS IR (Lift & Shift)**	**Medium**	Preserves row-by-row memor logic and custom assemblies without forcing a complete architectural rewrite.
Complex Merge Joins, Fuzzy Lookups, massive sorting operations, or loops	**Azure Databricks / Fabric Notebook**	**High**	Heavy transformation scale efficiently when pushed down into a distributed Apache Spark environment

SSIS Component Profile	Target Recommendation	Migration Complexity	Justification
iterating over millions of rows in memory.			rather than row-buffered i data integratio layers.

2. Automated Assessment Architecture

The tool uses a lightweight static parser layer combined with a frontier LLM scoring prompt to isolate logic from layout metadata:

A. Structural Feature Extraction Loop

The code parses the XML structure to evaluate elements using the following logic matrix:

```
# Core complexity classification logic def
evaluate_package_complexity(xml_nodes): score = 0
recommendation = "Native ADF Pipeline" flags = {
"has_script_tasks": xml_nodes.findall(".//{*}
Executables/{*}
Executable[@DTS:ExecutableType='Microsoft.ScriptTask']"),
"has_third_party": xml_nodes.findall(".//{*}
ObjectData[{*}customObject]"),
"has_heavy_transforms": xml_nodes.findall(".//{*}
property[@name='TransformationType']") } if
flags["has_script_tasks"] or
flags["has_third_party"]: score += 50 recommendation
= "Azure-SSIS IR (Lift & Shift)" if
flags["has_heavy_transforms"] and
len(flags["has_heavy_transforms"]) > 3: score += 30
recommendation = "Azure Databricks / Fabric
Notebook" return {"complexity_score": score,
"target_architecture": recommendation}
```

3. Automated Remediation Blueprint

Once routed, the tool provides the frontier model with specific transformation rules:

- **Variables & Expressions:** Convert `@[System::PackageName]` to `@pipeline().DataFactory` statements automatically.
- **Control Flow Branching:** Translate SSIS Precedence Constraints (Success, Failure, Completion) into native ADF activity dependency blocks (`dependsOn` array structures).

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